

PRACTICE EXERCISE

1. Which of the following are rational expressions?

I. $\frac{x^3 - 3x^2 + 2}{x^2 + 1}$ II. $\frac{z^3 - 3z^2}{2z + 3}$
 III. $\frac{x^2 - x + 2}{x + 3}$ IV. $\frac{x^3 + 3x^2 - 1}{x^2 + \sqrt{x - 1}}$

Select the correct answer using the codes given below

- (a) I, II and III (b) II, III and IV
 (c) All of these (d) None of these

2. The simplified form of $\frac{a+2}{a+3} - \frac{(a+1)}{(a+2)}$

(a) $\frac{1}{a^2 + 5a + 6}$ (b) $\frac{a+2}{a^2 + 5a + 6}$
 (c) 0 (d) $\frac{(a+2)^2}{a^2 + 5a + 6}$

3. $\left(\frac{x+1}{x^2-1} - \frac{2}{x}\right)$ expressed as a rational expression is

(a) $\frac{x^2-2}{x(x^2-1)}$ (b) $\frac{-(x-2)}{x(x-1)}$
 (c) $\frac{2(x+1)}{(x^3-1)}$ (d) $\frac{1}{x(x+1)}$

4. The expression $\frac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$ in lowest terms is

(a) $\frac{1}{x-7}$ (b) $(x-7)$ (c) $(x-2)$ (d) $\frac{1}{x-2}$

5. $\frac{a-c}{(a-b)(x-a)} + \frac{b-c}{(b-a)(x-b)}$ is equal to

(a) $\frac{b-a}{(x-b)(x-a)}$ (b) $\frac{x-c}{(x-a)(x-b)}$
 (c) $\frac{b-a}{(x-b)(x-c)}$ (d) None of these

6. The sum of the rational expression $\frac{x-3}{x^2+1}$ and its reciprocal is

(a) $\frac{x^3-3x^2+x-3}{x^4+3x^2-6x+10}$ (b) $\frac{x^4+3x^2-6x+10}{x^3-3x^2+x-3}$
 (c) $\frac{x^4-3x^2+6x+10}{x^3-3x^2+x-3}$ (d) $\frac{(x-1)^3}{(x+3)^2}$

7. The value of

$\frac{a}{(a-b)(a-c)} + \frac{b}{(b-c)(b-a)} + \frac{c}{(c-a)(c-b)}$
 (a) 2a (b) 0 (c) 2b (d) b-c

8. If $A = \frac{x-1}{x+1}$ and $B = \frac{x+1}{x-1}$, then $(A+B)^2$ is

(a) $\frac{4x^4+8x^2-4}{x^4-2x^2+1}$ (b) $\frac{4x^4+8x^2+4}{x^4-2x^2+1}$
 (c) $\frac{4x^4+8x^2+4}{x^4+2x+1}$ (d) None of these

9. The value of

$\frac{1}{(1-a)(1-b)} + \frac{a^2}{(1-a)(b-a)} - \frac{b^2}{(b-1)(a-b)}$ is
 (a) $1+a$ (b) $1-a^2$
 (c) $1-b^2$ (d) None of these

10. The value of $\frac{\left[\frac{x}{y} - \frac{y}{x}\right]\left[\frac{y}{z} - \frac{z}{y}\right]\left[\frac{z}{x} - \frac{x}{z}\right]}{\left[\frac{1}{x^2} - \frac{1}{y^2}\right]\left[\frac{1}{y^2} - \frac{1}{z^2}\right]\left[\frac{1}{z^2} - \frac{1}{x^2}\right]}$ is

(a) $x^2y^2z^2$ (b) $-x^2y^2z^2$
 (c) 1 (d) None of these

11. What is the simplified form of

$\left(\frac{x^2-3x+2}{x^3-8}\right) \div \left(\frac{x^2-9}{x^2+7x+12}\right) \times \left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right)$?
 (a) $\frac{x}{x-3}$ (b) $\frac{x-2}{x-3}$
 (c) $\frac{x}{x+3}$ (d) $\frac{x+3}{x+4}$

12. If $pq + qr + rp = 0$, then what is the value of

$\frac{p^2}{p^2-qr} + \frac{q^2}{q^2-rp} + \frac{r^2}{r^2-pq}$?
 (a) 0 (b) 1
 (c) -1 (d) 3

13. If $x + y + z = 0$, then what is the value of

$\frac{1}{x^2+y^2-z^2} + \frac{1}{y^2+z^2-x^2} + \frac{1}{z^2+x^2-y^2}$?
 (a) $\frac{1}{x^2+y^2+z^2}$ (b) 1
 (c) -1 (d) 0

14. If $a = \frac{1+x}{2-x}$, then what is $\frac{1}{a+1} + \frac{2a+1}{a^2-1}$ equal to?

(a) $\frac{(1+x)(2+x)}{2x-1}$ (b) $\frac{(1-x)(2-x)}{x-2}$
 (c) $\frac{(1+x)(2-x)}{2x-1}$ (d) $\frac{(1-x)(2-x)}{2x+1}$

15. If $\frac{1}{x+1} + \frac{2}{y+2} + \frac{1009}{z+1009} = 1$, then what is the value of $\frac{x}{x+1} + \frac{y}{y+2} + \frac{z}{z+1009}$?
- (a) 0 (b) 2 (c) 3 (d) 4

16. What is $\frac{x^2 - 3x + 2}{x^2 - 5x + 6} \div \frac{x^2 - 5x + 4}{x^2 - 7x + 12}$ equal to?
- (a) $\frac{x+3}{x-3}$ (b) 1 (c) $\frac{x+1}{x+3}$ (d) 2

17. If $x + y + z = 0$, then what is $\frac{xyz}{(x+y)(y+z)(z+x)}$ equal to (where, $x \neq -y, y \neq -z, z \neq -x$)?
- (a) -1 (b) 1
(c) $xy + yz + zx$ (d) None of these

PREVIOUS YEARS' QUESTIONS

18. What is $\frac{(x^2 + y^2)(x - y) - (x - y)^3}{x^2y - xy^2}$ equal to? ☑ 2013 II
- (a) 1 (b) 2 (c) 4 (d) -2
19. What is $\frac{1}{a-b} - \frac{1}{a+b} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8}$ equal to? ☑ 2014 II
- (a) $a+b$ (b) $a-b$ (c) 1 (d) 0
20. If $a^2 - by - cz = 0$, $ax - b^2 + cz = 0$ and $ax + by - c^2 = 0$, then the value of $\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z}$ will be ☑ 2016 I
- (a) $a+b+c$ (b) 3 (c) 1 (d) 0

ANSWERS

1	a	2	a	3	b	4	c	5	b	6	b	7	b	8	b	9	d	10	b
11	a	12	b	13	d	14	c	15	b	16	b	17	a	18	b	19	d	20	c

HINTS AND SOLUTIONS

1. (a) Here, $\frac{x^3 + 3x^2 - 1}{x^2 + \sqrt{x-1}}$ is not a rational expression, since the denominator is not a polynomial.

2. (a) We know $\frac{a+2}{a+3} - \frac{(a+1)}{(a+2)}$

$$= \frac{(a+2)^2 - (a+1)(a+3)}{(a+3)(a+2)}$$

$$= \frac{a^2 + 4a + 4 - (a^2 + 4a + 3)}{a^2 + 5a + 6}$$

$$= \frac{1}{a^2 + 5a + 6}$$

3. (b) Here, $\left(\frac{x+1}{x^2-1} - \frac{2}{x}\right) = \frac{1}{x-1} - \frac{2}{x}$

$[\because x^2 - 1 = (x+1)(x-1)]$

$$= \frac{x-2x+2}{x(x-1)} = \frac{-(x-2)}{x(x-1)}$$

4. (c) We know that,

$$\frac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$$

$$= \frac{(x-1)(x-2)(x^2-7x-2x+14)}{(x-7)(x^2-2x-x+2)}$$

$$= \frac{(x-1)(x-2)(x-7)(x-2)}{(x-7)(x-2)(x-1)} = x-2$$

5. (b) Here, $\frac{a-c}{(a-b)(x-a)} + \frac{b-c}{(b-a)(x-b)}$

$$= \frac{(a-c)(x-b) - (b-c)(x-a)}{(a-b)(x-a)(x-b)}$$

$$= \frac{ax-ab-xc+bc-(bx-ab-cx+ac)}{(a-b)(x-a)(x-b)}$$

$$= \frac{ax+bc-bx-ac}{(a-b)(x-a)(x-b)}$$

$$= \frac{x(a-b)-c(-b+a)}{(a-b)(x-a)(x-b)}$$

$$= \frac{(x-c)(a-b)}{(a-b)(x-a)(x-b)} = \frac{(x-c)}{(x-a)(x-b)}$$

6. (b) Here, reciprocal of $\frac{x-3}{x^2+1}$ is $\frac{x^2+1}{(x-3)}$

So,

$$\frac{x-3}{x^2+1} + \frac{x^2+1}{x-3} = \frac{(x-3)^2 + (x^2+1)^2}{(x^2+1)(x-3)}$$

$$= \frac{x^2+9-6x+x^4+1+2x^2}{x^3-3x^2+x-3}$$

$$= \frac{x^4+3x^2-6x+10}{x^3-3x^2+x-3}$$

7. (b) We know that,

$$\frac{a}{(a-b)(a-c)} + \frac{b}{(b-c)(b-a)} + \frac{c}{(c-a)(c-b)}$$

$$= \frac{-a}{(a-b)(c-a)} - \frac{b}{(b-c)(a-b)} - \frac{c}{(c-a)(b-c)}$$

$$= \frac{(-a)(b-c) - b(c-a) - c(a-b)}{(a-b)(b-c)(c-a)}$$

$$= \frac{-ab+ac-bc+ab-ac+bc}{(a-b)(b-c)(c-a)}$$

$$= \frac{0}{(a-b)(b-c)(c-a)} = 0$$

8. (b) Here,

$$(A+B)^2 = \left[\left(\frac{x-1}{x+1} \right) + \left(\frac{x+1}{x-1} \right) \right]^2$$

$$= \left[\frac{(x-1)^2 + (x+1)^2}{(x+1)(x-1)} \right]^2 = \left[\frac{2(x^2+1)}{x^2-1} \right]^2$$

$[\because (a+b)^2 + (a-b)^2 = 2(a^2+b^2)]$

$$= \frac{4(x^4+2x^2+1)}{x^4-2x^2+1} = \frac{4x^4+8x^2+4}{x^4-2x^2+1}$$

9. (d) Given,

$$\frac{1}{(1-a)(1-b)} + \frac{a^2}{(1-a)(b-a)} - \frac{b^2}{(b-1)(a-b)}$$

$$= \frac{(b-a) + a^2(1-b) - b^2(1-a)}{(1-a)(1-b)(b-a)}$$

$$\begin{aligned}
 &= \frac{b-a+a^2-a^2b-b^2+ab^2}{(1-a)(1-b)(b-a)} \\
 &= \frac{(b-a)+(a^2-b^2)+ab(b-a)}{(1-a)(1-b)(b-a)} \\
 &= \frac{(b-a)-(a+b)(b-a)+ab(b-a)}{(1-a)(1-b)(b-a)} = \frac{1-a-b+ab}{1-b-a+ab} = 1
 \end{aligned}$$

10. (b) Given, $\left[\frac{x}{y} - \frac{y}{x}\right] \left[\frac{y}{z} - \frac{z}{y}\right] \left[\frac{z}{x} - \frac{x}{z}\right]$

$$\begin{aligned}
 &= \left[\frac{1}{x^2} - \frac{1}{y^2}\right] \left[\frac{1}{y^2} - \frac{1}{z^2}\right] \left[\frac{1}{z^2} - \frac{1}{x^2}\right] \\
 &= \left[\frac{x^2-y^2}{x^2y^2}\right] \left[\frac{y^2-z^2}{z^2y^2}\right] \left[\frac{z^2-x^2}{x^2z^2}\right] \\
 &= \frac{(x^2-y^2)(y^2-z^2)(z^2-x^2)}{x^2y^2z^2} \\
 &= -\frac{(x^2-y^2)(y^2-z^2)(z^2-x^2)}{x^4y^4z^4} = -\frac{x^4y^4z^4}{x^4y^4z^4} = -x^2y^2z^2
 \end{aligned}$$

11. (a) Given, $\left(\frac{x^2-3x+2}{x^3-8}\right) \div \left(\frac{x^2-9}{x^2+7x+12}\right) \times \left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right)$

$$\begin{aligned}
 &= \left(\frac{x^2-3x+2}{x^3-8}\right) \times \frac{x^2+7x+12}{x^2-9} \times \left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right) \\
 &= \frac{(x-1)(x-2)}{(x-2)(x^2+4+2x)} \times \frac{(x+4)(x+3)}{(x-3)(x+3)} \times \frac{x(x^2+2x+4)}{(x-1)(x+4)} \\
 &= \frac{x}{x-3} \quad [\because a^3-b^3 = (a-b)(a^2+b^2+ab)]
 \end{aligned}$$

12. (b) Given, $pq+qr+rp=0$

$$\begin{aligned}
 \therefore \frac{p^2}{p^2-qr} + \frac{q^2}{q^2-rp} + \frac{r^2}{r^2-pq} \\
 &= \frac{p^2}{p^2+rp+pq} + \frac{q^2}{q^2+pq+qr} + \frac{r^2}{r^2+qr+rp} \\
 &= \frac{p^2}{p(p+r+q)} + \frac{q^2}{q(p+q+r)} + \frac{r^2}{r(p+q+r)} = \frac{p+q+r}{p+q+r} = 1
 \end{aligned}$$

13. (d) Given, $x+y+z=0 \Rightarrow x+y=-z$
 On squaring both sides, we get $x^2+y^2+2xy=z^2$
 $x^2+y^2-z^2=-2xy$
 Similarly, $y^2+z^2-x^2=-2yz$ and $z^2+x^2-y^2=-2zx$

$$\begin{aligned}
 \therefore \frac{1}{x^2+y^2-z^2} + \frac{1}{y^2+z^2-x^2} + \frac{1}{z^2+x^2-y^2} \\
 &= \frac{1}{-2xy} + \frac{1}{-2yz} + \frac{1}{-2zx} = \frac{1}{2} \left(\frac{z+x+y}{xyz} \right) = 0
 \end{aligned}$$

14. (c) Given, $a = \frac{1+x}{2-x}$ So, $\frac{1}{a+1} + \frac{2a+1}{a^2-1} = \frac{3a}{a^2-1} = \frac{3\left(\frac{1+x}{2-x}\right)}{\left(\frac{1+x}{2-x}\right)^2-1}$

$$\begin{aligned}
 &= \frac{3(1+x)(2-x)}{1+x^2+2x-(4+x^2-4x)} = \frac{3(1+x)(2-x)}{6x-3} = \frac{(1+x)(2-x)}{(2x-1)}
 \end{aligned}$$

15. (b) Given, $\frac{1}{x+1} + \frac{2}{y+2} + \frac{1009}{z+1009} = 1$

$$\begin{aligned}
 \Rightarrow \frac{1}{x+1} - 1 + \frac{2}{y+2} - 1 + \frac{1009}{z+1009} - 1 &= 1-3 \\
 \Rightarrow -\frac{x}{x+1} - \frac{y}{y+2} - \frac{z}{z+1009} &= -2 \\
 \therefore \frac{x}{x+1} + \frac{y}{y+2} + \frac{z}{z+1009} &= 2
 \end{aligned}$$

16. (b) Given, $\frac{x^2-3x+2}{x^2-5x+6} \div \frac{x^2-5x+4}{x^2-7x+12}$

$$\begin{aligned}
 &= \frac{x^2-3x+2}{x^2-5x+6} \times \frac{(x^2-7x+12)}{(x^2-5x+4)} \\
 &= \frac{(x-1)(x-2)}{(x-3)(x-2)} \times \frac{(x-4)(x-3)}{(x-4)(x-1)} = 1
 \end{aligned}$$

17. (a) Given, $x+y+z=0$

$$\begin{aligned}
 \Rightarrow x+y=-z, y+z=-x \text{ and } z+x=-y \\
 \therefore \frac{xyz}{(x+y)(y+z)(z+x)} = \frac{xyz}{(-z)(-x)(-y)} = \frac{xyz}{-xyz} = -1
 \end{aligned}$$

18. (b) Given, $\frac{(x^2+y^2)(x-y)-(x-y)^3}{x^2y-xy^2}$

$$\begin{aligned}
 &= \frac{x^3+xy^2-x^2y-y^3-(x^3-y^3-3x^2y+3xy^2)}{x^2y-xy^2} \\
 &= \frac{x^3+xy^2-x^2y-y^3-x^3+y^3+3x^2y-3xy^2}{x^2y-xy^2} \\
 &= \frac{2x^2y-2xy^2}{x^2y-xy^2} = \frac{2(x^2y-xy^2)}{x^2y-xy^2} = 2
 \end{aligned}$$

19. (d) $\frac{1}{a-b} - \frac{1}{a+b} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8}$

$$\begin{aligned}
 &= \frac{(a+b)-(a-b)}{(a-b)(a+b)} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{2b}{a^2-b^2} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{2b(a^2+b^2)-2b(a^2-b^2)}{(a^2-b^2)(a^2+b^2)} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{4b^3}{a^4-b^4} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{4b^3(a^4+b^4)-4b^3(a^4-b^4)}{(a^4-b^4)(a^4+b^4)} - \frac{8b^7}{a^8-b^8} = 0
 \end{aligned}$$

20. (c) Given, $a^2-by-cz=0$... (i)
 $ax-b^2+cz=0$... (ii)
 and $ax+by-c^2=0$... (iii)

On adding Eqs. (i), (ii) and (iii), we get

$$\begin{aligned}
 x &= \frac{b^2-a^2+c^2}{2a} \Rightarrow \frac{x}{a+x} = \frac{b^2-a^2+c^2}{a^2+b^2+c^2} \\
 \text{Similarly, } \frac{y}{b+y} &= \frac{a^2-b^2+c^2}{a^2+b^2+c^2} \text{ and } \frac{z}{c+z} = \frac{a^2+b^2-c^2}{a^2+b^2+c^2} \\
 \therefore \frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z} &= \frac{(b^2-a^2+c^2)+(a^2-b^2+c^2)+(a^2+b^2-c^2)}{a^2+b^2+c^2} = 1
 \end{aligned}$$

LINEAR EQUATIONS

Generally (3-5) questions have been definitely asked from this chapter in CDS examination. Beside this, the linear equations are also applicable in solving various word problems related to number system, mensuration, time and work, etc.



A linear equation is an equation for a straight line.

So, the equation which has degree 1, i.e., which has linear power of the variables, is called a linear equation.

LINEAR EQUATIONS IN ONE VARIABLE

A linear equation in one variable is an equation which can be written in the form of $ax + b = 0$, or $ax = c$, where a , b , c are real numbers with $a \neq 0$.

e.g. $5x + 8 = 9 - x$, $\frac{2}{3}y + 7 = \frac{y}{2}$ and $t + 3t = 9 - t$ are linear equations in one variable.

Solving a linear Equation in One Variable

1. Simplify both sides of the equation.
2. Use the addition and subtraction properties to get all variables of terms on the LHS and all constant terms on the RHS.
3. Simplify and divide both sides of the equation by the coefficient of the variable.

EXAMPLE 1. Solve $2(x - 3) - (5 - 3x) = 3(x + 1) - 4(2 + x)$.

a. 1

b. -1

c. 0

d. 3

Sol. a. $2(x - 3) - (5 - 3x) = 3(x + 1) - 4(2 + x) \Rightarrow 2x - 6 - 5 + 3x = 3x + 3 - 8 - 4x$
 $\Rightarrow 5x - 11 = -x - 5 \Rightarrow 6x = 6 \Rightarrow x = 1$, Hence, the value of x is 1.

LINEAR EQUATION IN TWO VARIABLES

An equation which can be put in the form $ax + by + c = 0$, where a , b and c are real numbers and a , b both are not zero, is called a linear equation in two variables.

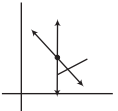
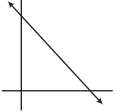
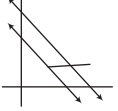
e.g. $2x + 3y = 5$, $\sqrt{2}x + \sqrt{3}y = 0$ and $2a + 3b = 0$ are linear equations in two variables.

Pair of Linear Equations

Two linear equations in the two same variables are called pair of linear equations in two variables. The general form of pair of linear equations in two variables x and y is $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ where, a_1, a_2, b_1, b_2 and c_1, c_2 are all real numbers and $a_1^2 + b_1^2 \neq 0, a_2^2 + b_2^2 \neq 0$.

Consistency Pair of linear Equations

A pair of linear equations which has no solution is called an **inconsistent pair** of linear equations. A pair of linear equations which has atleast one solution is called a **consistent pair** of linear equations. Let us consider two linear equations as $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$. Pair of linear equations in two variables will represent two straight lines, both to be considered together. The table given below represents the three possibilities.

Ratios	Consistency	Graphical Representation	Algebraic interpretation
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Consistent	 Intersecting lines.	Exactly one solution (unique).
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Dependent (consistent)	 Coincident lines.	Infinitely many solutions.
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Inconsistent	 Parallel lines.	No solution.

HOMOGENEOUS PAIR OF LINEAR EQUATIONS

The pair of linear equations $a_1x + b_1y = 0, a_2x + b_2y = 0$ has

1. A unique solution, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ and the solution is $x = 0, y = 0$.
2. An infinite number of solutions, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$.

This pair of linear equations is always consistent.

EXAMPLE 2. The values of k for which the system $kx + 2y = 5$ and $3x + y = 1$ has

- (i) unique solution (ii) no solution are
- a. $k \neq 6$ and $k = 6$ b. $k \neq 3$ and $k = 3$
- c. $k \neq 2$ and $k = 2$ d. None of these

Sol. a. Given equations are, $kx + 2y - 5 = 0$ and

$$3x + y - 1 = 0$$

On comparing the given equations, with standard form of equations

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0$$

$$\text{We have, } a_1 = k, b_1 = 2, c_1 = -5$$

$$\text{and } a_2 = 3, b_2 = 1, c_2 = -1$$

(i) Here, the equations have a unique solution, if

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\text{i.e. if } \frac{k}{3} \neq \frac{2}{1} \Rightarrow k \neq 6$$

(ii) The equations have no solution, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\therefore \frac{k}{3} = \frac{2}{1} \Rightarrow k = 6$$

EXAMPLE 3. The value of k for which the system of equations $kx - y = 2, 6x - 2y = 3$ has infinitely many solutions is

a. 3

b. 4

c. 12

d. No such value of k exists.

Sol. d. Given equations are,

$$kx - y - 2 = 0 \text{ and } 6x - 2y - 3 = 0$$

On comparing the given equations, with standard form of equations

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0$$

$$\text{We have, } a_1 = k, b_1 = -1, c_1 = -2$$

$$\text{and } a_2 = 6, b_2 = -2, c_2 = -3$$

Here, the equations have infinite number of solutions, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\text{But, } \frac{a_1}{a_2} = \frac{k}{6}, \frac{b_1}{b_2} = \frac{-1}{-2} = \frac{1}{2} \text{ and } \frac{c_1}{c_2} = \frac{-2}{-3} = \frac{2}{3}$$

$$\text{So, } \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Thus, the system has no such value of k , for which the given system has infinitely many solutions.

Algebraic Methods of Solutions

Different algebraic methods for solving pair of linear equations are discussed below in detail.

1. Substitution Method

In this method, we substitute the value of one variable in terms of other variable to solve the pair of linear equations, so this method is called substitution method.

Steps used in this method are given below.

Step I Find the value of one variable x (or y) in terms of other variable i.e. y (or x) from an equation.